CHECKLIST
SMALL CRAFT - HYDRAULIC STEERING SYSTEMS
Ref.: EN ISO 10592:2022 (ISO 10592:2022)

Checklist_Evaluation_Module B_G en260703

Watercraft manufacturer:	
Watercraft model name:	

	Subject to check	Clause	Requirements	Checked ?
1	If craft with OB engine, steering system is complete from the control element to the mechanical interface for connection of the drag link supplied with the outboard engine. Or it provides an alternative means to connect the output device to the engine so that the loading magnitude and offset are consistent with the steering arm's intended purpose.	4.1.a	[Yes / NA ?]	
2	If non-OB craft, the steering system shall be complete from the control element to the output connection point on the steerable device.	4.1.b	[Yes / NA ?]	
3	If integrity threaded of fasteners affects operation of the remote hydraulic steering system so that separation or loss of the fastener would cause total loss of steering without warning shall be provided with a locking means. This requirement does not apply to hydraulic fittings.	4.2	[Yes / NA ?]	
4	If integrity of threaded fasteners affects operation of the steering system so that separation or loss of the fasteners can cause total loss of steering without warning, and that can be expected to be disturbed by installation or adjustment procedures, shall be referenced by instructions for correct assembly, and: - shall be locked by a device whose presence is determined by visual inspection, or by feel, following assembly, or - shall incorporate integral locking means, provided the fastener cannot be omitted or substituted without making the system inoperable. This requirement does not apply to hydraulic fittings. Self-locking nuts with plastic inserts that create mechanical plastic interference meet the above stated requirements.	4.3	[Yes / NA ?]	
5	Loose lock washers, distorted thread nuts or separately applied adhesives shall not be used.	4.4	[Yes / NA ?]	
6	Devices that use plain threaded jam nuts to permit adjustments shall be designed so that total separation of parts, or total loss of steering, will not occur should they loosen.	4.5	[Yes / NA ?]	
7	Connection fittings, including quick-disconnect fittings relying only upon a spring or springs to maintain the connection, shall not be used.	4.6	[Yes / NA ?]	

compliant: Yes or √ not applicable: NA follow up on variation report: Rpt or X

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8	Hydraulic lines shall be of sufficient length to permit installation of the output device for single or multiple engine installations on those craft designed for multiple outboard engine installations.	8.1	[Yes ?]	
9	Hydraulic lines shall accommodate the full range of intended travel without interference with the mechanical interface requirements of an outboard engine steering system including the full range of the engine tilt and trim.	8.2	[Yes ?]	
10	Hydraulic lines shall be routed so that the ambient temperature in the space does not exceed the operating temperature range specified for the hydraulic lines used.	8.3	[Yes ?]	
11	There shall be no joints or connections in hydraulic lines directly over exhaust system components or high temperature manifolds.	8.3	[Yes ?]	
12	Hydraulic lines shall be installed with as few bends as practicable. Bends shall have as large a radius as practicable, and the radius shall not be smaller than the line manufacturer’s recommended minimum.	8.4	[Yes ?]	
13	The steering manufacturer recommended bend radius of the hydraulic line shall be provided on product or in the installation manual.	8.4	[Yes ?]	
14	Hydraulic lines shall be selected and routed to avoid any stretching, crushing, restricted movement, kinking, or chafing.	8.5	[Yes ?]	
15	The installation shall be carried out following the directions of the manufacturers of the system. Hydraulic lines shall be supported by clips, straps or other means to prevent chafing or vibration damage. The clips, straps or other devices shall be corrosion-resistant and shall be designed to prevent cutting, abrading or damage to the lines, and shall be compatible with hydraulic line materials.	8.5	[Yes ?]	
16	Hydraulic lines and component ports/fittings should be capped/plugged until the hydraulic lines and components are fully interconnected to prevent contamination.	8.5	[Yes ?]	
17	Hydraulic lines shall be routed to avoid any contact with sharp edges/screws.	8.6	[Yes ?]	
18	Outboard steering systems, where the hydraulic lines must reciprocate with the steering cylinder, shall use flexible hydraulic hoses, and shall not use rigid tubing at the cylinder connection.	8.7	[Yes ?]	
19	If the hydraulic lines pass through the side of an outboard engine well below the downflooding height, as defined in ISO 12217-series, the opening shall meet the degree of watertightness requirements specified in this ISO standard.	8.8	[Yes ?]	
20	Ball joints used to connect the steering system to the steerable device shall be installed so that total loss of steering does not occur if the ball separates axially from its socket. Note that a flat washer larger than the socket bore can meet this requirement.	8.9	[Yes / NA ?]	
21	Steering wheels and helm shafts shall be selected to fit each other. Current fit configurations are shown in Figure 9.	8.10	[Yes ?]	
22	Hydraulic helm is permanently marked:		[Yes ?]	
23	- largest diameter and dish, visible when helm installed when wheel is removed	7.4	[Yes ?]	

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24	- system proof pressure, visible at the front of the helm with the steering wheel installed or at the back of the helm adjacent to the hydraulic line connection.	7.4	[Yes ?]	
25	- a reference to this standard	7.4	[Yes ?]	
26	- name/trademark of manufacturer	7.4	[Yes ?]	
27	- model type	7.4	[Yes ?]	
28	- year of production.	7.4	[Yes ?]	
29	Cylinders (output device) permanently marked:			
30	- component pressure proof	7.5	[Yes ?]	
31	- a reference to this standard	7.5	[Yes ?]	
32	- a name or trademark of the manufacturer	7.5	[Yes ?]	
33	- year of production	7.5	[Yes ?]	
34	The component manufacturer's requirement for the hydraulic fluid shall be permanently and legibly displayed, adjacent to the filling location of the system or on the cap.	7.6	[Yes ?]	

The following questions shall be filled in by the watercraft manufacturer and appropriate documentation shall be submitted to the inspector for verification.

Subject to check		Clause	Requirements	Checked ?
35	For operating the temperature range for all materials used in the construction of the system and its accessories shall be capable of operating from –20 °C to +80 °C.	4.7	[Yes ?]	
36	Hydraulic system components shall not be installed in areas where the operating temperature exceeds +80 °C.	4.7	[Yes ?]	
37	For storage all materials used in the construction of the system and its accessories shall be capable of withstanding an ambient temperature range of –40 °C to +85 °C.	4.8	[Yes ?]	
38	All components including, but not limited to, hydraulic lines and fittings, and input and output devices shall be marked and selected to have a component proof pressure rating not less than the proof pressure rating on the hydraulic helm as indicated by the manufacturer of the helm.	4.9	[Yes ?]	
39	Components shall have a burst pressure that is not less than the system design peak pressure throughout the operating temperature range and expected burst pressure variation due to manufacture, installation, environmental exposure and in use loading, or two times component proof pressure, whichever is greater.	4.10	[Yes ?]	
40	Hydraulic lines and fittings shall be selected in accordance with the steering equipment manufacturers' instructions.	4.11	[Yes ?]	
41	If applicable, hydraulic quick connect fittings whose integrity affects operation of the system so that separation or loss of the connection would cause total loss of steering without warning shall incorporate a two stage integral locking means for connection integrity.	4.11	[Yes / NA ?]	

compliant: Yes or √ not applicable: NA follow up on variation report: Rpt or X

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42	Hydraulic steering systems, including systems with pressure relief devices, shall comply with the following test to ensure that movement after a relief event is controlled: - apply an impulse load of at least the system design peak pressure followed directly with at least one-half system design peak pressure for a duration of at least one-half second; - the load shall be applied to the steerable device and resisted by the steering system; - the load shall be applied within 13 degrees of steering centre; - the system shall not have more than 17 degrees of steering movement.	4.12	[Yes ?]	
43	Steering systems shall cause the steerable device to turn on its axis at its rate of steering response when no greater than 4 % of the full range of the steering movement of the steering wheel or control element.	4.13	[Yes ?]	
44	Steering systems shall not cause the operator to re-grip the steering wheel or handgrip more frequently than once every 30 s due to position drift of 1/4 turn or more of the steering wheel or handgrip relative to the position of the steerable device.	4.14	[Yes ?]	
45	Component interfaces and hardware shall be capable of withstanding the forces generated by the system operating at the system design peak pressure.	4.15	[Yes ?]	
46	If applicable, in multiple engine installations that are not mechanically connected, sudden loss of steering synchronization shall be prevented. Series plumbing of steering components meets this requirement.	4.16	[Yes / NA ?]	
47	When equipped with the largest diameter Ds and the deepest dish of the steering wheel for which the helm is rated, all steering components shall be capable of meeting the applicable test requirements specified in Clause 9.	4.17	[Yes ?]	
48	Materials used in remote hydraulic steering systems shall be galvanically compatible or suitably plated to minimize corrosion.	5.1	[Yes ?]	
49	If copper-base alloys are used, they are separated from aluminium with a galvanic barrier, such as 300 series stainless steel or equivalent, or shall be protected from exposure.	5.2	[Yes / NA ?]	
50	Metallic steering components that are at or below the waterline in the light craft condition, (see ISO 12217-seriew) these are cathodic protected or galvanically isolated.	5.3	[Yes / NA ?]	
51	Materials used in remote hydraulic steering systems shall be resistant to deterioration by the specified hydraulic fluid and by other liquids or compounds with which the material can come in contact under normal marine services, e.g. grease, lubricating oil, common bilge solvents, and salt and fresh water.	5.4	[Yes ?]	
52	Plastics and elastomers that can be exposed to sunlight shall be designed to resist degradation by ultraviolet radiation.	5.5	[Yes / NA ?]	
53	The hydraulic fluid shall be non-flammable or have a flash point of 160 °C or over.	5.6	[Yes ?]	

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54	If OB engine craft, the steering stops do permit at least 30° of angular movement either side of centre.	5.7	[Yes / NA ?]	
55	If OB engine, it shall: a) incorporate an integrated steering system or meet the applicable dimensional requirements indicated in Figure 3 and Figure 4, and b) provide space for the connection of the steering components as indicated in Figure 5, Figure 6 and Figure 7.	5.8	[Yes / NA ?]	
56	If OB engine(s), the steering system is designed so that, with any combination of engine turn and tilt, there shall be no damaging interference between the engine(s), its accessories and the craft. Appropriate written information and installation instructions shall be provided, clearly indicating the type of steering system(s) that should be used with outboard engine(s).	5.9	[Yes / NA ?]	
57	Engine-mounted steering systems shall incorporate the dimensional requirements indicated in Figure 3 and Figure 4.	7.1	[Yes / NA ?]	
58	Steering systems with the end of the output device co-axial with the engine-mounting steering tube shall meet the mechanical interface requirements of ISO 8848:2022.	7.2	[Yes / NA ?]	
59	Craft-mounted steering systems and engine-mounted steering systems for outboard engine installations shall meet the requirements of 6.3.	7.3	[Yes / NA ?]	
60	The tests in 9.2 are intended to verify the integrity and function of each steering system as installed in the craft. Tests shall be performed upon original installation, when system component changes are made, and when servicing results in the disconnection/reconnection of mechanical or hydraulic interfaces.	9.1.1	[Yes ?]	
61	Each installed steering system shall withstand a proof pressure test at each hard over position without leakage, disconnection, or permanent deflection of system components. The tests shall be conducted as follows: a sufficient tangential force shall be applied to the steering wheel rim to cause the hydraulic steering system including the output device to experience system proof pressure for a minimum duration of 60 s, during which time all hydraulic, mechanical and component interfaces shall not leak.	9.1.2	[Yes ?]	
62	These tests are intended to qualify the application of a steering system installed in a particular model of craft rigged with intended output devices.	9.2.1	[Yes ?]	
63	Installed steering systems shall complete two full cycles from hard over to hard over during which all moving components are inspected to confirm that no interference or restriction of moving components is present through the full range of travel. For outboard engine and sterndrive installations, the requirements of 6.3 shall be confirmed by testing under all combinations of trim, tilt, elevation, and steering angle.	9.2.2	[Yes ?]	
64	During this test, installed systems shall demonstrate that no interference between the output device, the steerable device, tiebar, transom, or adjustable engine lift plate, engine well, or other surfaces occur.	9.2.2.1	[Yes ?]	
	No stretching, crushing, restricted movement of hydraulic lines, kinking of lines, or chafing of lines against bulkhead/engine well entry points or any other contact points shall occur.	9.2.2.2	[Yes ?]	

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	Installed steering systems shall be tested for conformance to the steering response requirements of 4.13.	9.2.3	[Yes ?]	
	Installed steering systems shall be tested for their conformance to the requirements of 4.14. This test shall verify the ability to maintain course.	9.2.4	[Yes ?]	

	Instructions/Warnings to be included in the owner's manual			
33	Operating instructions.	10	[Yes ?]	
34	Filling and bleeding procedures.	10	[Yes ?]	
35	In the event of failure, alternative means of operation.	10	[Yes / NA ?]	
36	Maintanance procedures.	10	[Yes ?]	
37	Trouble correction guidlines with warnings.	10	[Yes ?]	
38	Specifications for hydraulic fluid.	10	[Yes ?]	
39	System diagram of typical installation.	10	[Yes ?]	
	Part number and designation of frequent and easily replaceable components parts.	10	[Yes ?]	
40	Largest dish and diameter of a steering wheel that can be used on the steering helm.	10	[Yes ?]	

Comments: